Exercise list R8 — Corporate governance, control rights and investor activism

Answer key (full derivations in `lista-r8-governance-activism-solutions.pdf`)

1. Management opposes a hostile offer (Ross CQ 1.14)

Premium $35/25 - 1 = 40\%$, \$10 per share lost if the bid fails. Resisting serves shareholders only if management credibly expects a stand-alone value above \$35 or uses resistance to extract a higher bid; an immediate rejection fits the agency explanation (jobs and control benefits). Not necessarily in the shareholders' interest.

2. Ownership structure and institutional investors (Ross CQ 1.15)

Germany/Japan: large bank and corporate blocks monitor managers, so the manager–shareholder problem is *less* serious, but controlling-vs-minority conflicts and creditor-oriented (low-risk) decisions are more serious. Active institutions reduce free riding and strengthen shareholder control, adding the agency problem of the funds' own managers.

3. Determinants of ownership concentration (G&T 18.2)

More concentration with large monitoring gains (hard-to-monitor, uncertain firms), large private control benefits (intangibles), founder incentives/signalling/taxes, weak legal protection of minorities and permissive rules for institutions; less with high costs of undiversified risk (large, risky firms). Example 18.1: \$60m buys 20% of a \$300m firm; a \$60m gain is 20% over five years, $(1.2)^{1/5}-1 = 3.71\%$ a year — not worth the lost diversification.

4. A law against blockholders' private benefits (G&T 18.5)

Blockholder monitors iff $\alpha\Delta V + b_p \geq c$. With $\alpha = 0.1$, $\Delta V = 50$, $c = 8$: $5 < 8$ without private benefits; $5 + 4 = 9 \geq 8$ with $b_p = 4$ (minimum 3); others gain $45 - 4 = 41$. The law cuts b_p, so monitoring falls; minorities face less diversion but less monitoring: effect on value ambiguous.

5. Board control and debt ratios (G&T 18.4)

Debt: firm 2 > firm 1 > firm 3. Firm 2's board uses leverage to discipline investment it cannot watch (Example 18.3: senior debt $F \in (25, 75]$ funds the project in the good/medium states, $175 - F \geq 100$, and blocks it in the bad state, $125 - F < 100$); firm 1 sets debt by the tax/distress trade-off; firm 3's managers avoid debt (Result 18.6).

6. Proxy contests (Amaro de Matos 8.2.6 #10)

With $E_2 + K_2 \geq E_1$: worst in control minus best out of control $= K_2 - \alpha_j(E_1 - E_2) \geq (1 - \alpha_j)(E_1 - E_2) > 0$, so self-voting is dominant for any signal. Example $0.3(80) + 25 = 49 > 0.3(100) = 30$; with $K_2 = 5$, $29 < 30$ and it fails. Matches p. 278.

7. The large shareholder (Amaro de Matos 8.1.5 #5)

(a) $\pi^* = \min\{\pi : \pi \geq E[Z \mid Z \geq (1 - 2\alpha)\pi + 2c]\}$ falls with α. **(b)** $Z^c = (1 - 2\alpha)\pi^* + 2c$ falls. **(c)** $V = q + I^* \int_{Z^c} Z\, dF$ rises: lower Z^c and higher I^* ($c'(I^*) = \frac{1}{2}\int_{Z^c}(Z - Z^c)dF$). Uniform, $c = 0$: $\pi^* = (1 + Z^c)/2$ and $Z^c = (1 - 2\alpha)\pi^*$ give $Z^c = (0.5 - \alpha)/(0.5 + \alpha)$, $\pi^* = 1/(1 + 2\alpha)$;

$d\pi^*/d\alpha = -2/(1+2\alpha)^2$, $dZ^c/d\alpha = -1/(0.5+\alpha)^2$. $\alpha = 0.1$: $Z^c = 2/3$, $\pi^* = 0.833$; $\alpha = 0.2$: $Z^c = 3/7$, $\pi^* = 0.714$. Matches pp. 271–272.

8. Investor control can hurt financing (Tirole 10.5)

(i) $p_H(R - B/\Delta p) \geq I - A$. **(ii)** $p_H R_b \geq p_L R_b + B$ and $p_H R_b \geq (p_L + \nu)R_b + B - \gamma$; at $R_b = B/\Delta p$ the second has slack $\gamma - \nu B/\Delta p$: unchanged income if $\nu B/\Delta p \leq \gamma$, else $R_b = (B-\gamma)/(\Delta p - \nu)$ and income falls. **(iii)** $(\Delta p - \nu)R_b \geq B$: $p_H[R - B/(\Delta p - \nu)] \geq I - A$; NPV also falls by γ. Example ($p_H = 0.8$, $\Delta p = 0.3$, $B = 0.6$, $R = 5$, $\nu = 0.1$): 2.4 (or 2.2 if $\gamma = 0.15$) vs. 1.6.

9. Limited vs. extended investor control (Tirole 10.7)

Limited iff $(p_H + \tau_A)(R - B/\Delta p) \geq I - A$; extended iff only $(p_H + \tau_B)(R - B/\Delta p) \geq I - A$; otherwise no financing. $\tau_A R \geq \gamma_A$ rules out entrepreneur control. Example: $A \geq 1.4$ limited, $0.6 \leq A < 1.4$ extended (NPVs 0.9, 0.8).

10. Control and incentives (Tirole 10.6)

(i) $p_H(R - B/\Delta p) \geq I - A$. **(ii)** $\mathcal{P}_{EC} = p_H(R - B/\Delta p)$ (she never takes the action); $\mathcal{P}_{IC} = (p_H + \tau_H)[R - B/(\Delta p + \Delta\tau)]$. Investor control helps always if $\Delta\tau \geq 0$; if $\Delta\tau < 0$ only when $\Delta p + \Delta\tau > B(p_H + \tau_H)/(\tau_H R + p_H B/\Delta p)$. Example: threshold $\Delta\tau^* = -0.026$; $\Delta\tau = -0.05$ gives $3.25 < 3.6$.

11. Public debt, bank debt, junk bonds (Tirole 9.1)

$A_3 = I - p_H(R - B/\Delta p)$, $A_2 = I + c - p_H(R - b/\Delta p)$, $A_1 = I - p_L R$; $A_3 - A_2 = p_H(B-b)/\Delta p - c > 0$, $A_2 - A_1 = c - [\Delta p R - p_H b/\Delta p] > 0$. Utilities $p_H R - I > p_H R - I - c > p_L R + B - I > 0$ (second gap $= \Delta p R - c - B$). Example: $A_1 = 0.5 < A_2 = 0.6 < A_3 = 0.8$. Matches pp. 606–607.

12. Large minority blockholder (Tirole 10.3)

Aligned blockholder ($\xi \leq (\tau + \mu)s_2 R$): friendly action in all states. Biased: agreement (state 2) $\Rightarrow$ friendly; disagreement (states 1, 3) $\Rightarrow$ majority takes $\max\{V_E, V_L, 0\}$ with $V_E = -\beta\mu + (1-\beta)(1-\kappa)\tau$, $V_L = \beta\tau - (1-\beta)(1-\kappa)\mu$ (the book drops $(1-\beta)$ in V_E). If both are negative, both informed parties prefer to agree on the friendly action: friendly everywhere. (ii) In the collusive state both push the hostile action; agreement is adopted only if $\pi_2\tau \geq \pi_4\mu$.

13. Uncertain managerial horizon (Tirole 10.8)

$U_b = (1-\lambda)[(p_H + \tau x)R - \gamma x] + \lambda(p_H + \tau y)(R - B/\Delta p) - I$ = social value minus the new manager's rent $\lambda[(p_H + \tau y)B/\Delta p - \gamma y]$; breakeven $(1-\lambda)(p_H + \tau x)(R - B/\Delta p) + \lambda(p_H + \tau y)(R - B/\Delta p) \geq I - A$. $y = 1$; $x = 0$ if $(1-\lambda)\rho_0 + \lambda\rho_0^+ \geq I - A$ (then invest iff $(1-\lambda)\rho_1 + \lambda\rho_0^+ \geq I$), else the smallest x meeting breakeven, if $\rho_0^+ \geq I - A$ and $U_b \geq 0$. Example: $x = 0.531$, $U_b = 0.035$.

14. Continuum of control rights (Tirole 10.9)

(i) $x = 1$ iff $g \leq [R + \mu(R - B/\Delta p)]\, t$, $R_b = B/\Delta p$. **(ii)** Lower $A \Rightarrow$ higher $\mu \Rightarrow$ steeper line: τ, γ rise. **(iii)** $t > 0$: investors if $x = 1$, entrepreneur if $x = 0$; $t < 0$: entrepreneur if $x = 1$, investors if $x = 0$; no renegotiation. **(iv)** $t^* = g/k$, $d\gamma/d\tau = g/t^*$, $d^2\gamma/d\tau^2 = g/[t^{*3}h(t^*)] > 0$ (example: 13 and 878.8).

15. Control over liquidation (Tirole 10.2)

(i) Continue iff $\rho_1 - U_b^0 \geq \mu L - (\mu - 1)\rho_0$, $\mu \geq 1$; $R_b = B/\Delta p$, L to investors. **(ii)** $\Omega^{SB} \subseteq \Omega^{FB}$ (lines pivot at $(\rho_0, \rho_1 - \rho_0)$); shrinks as A falls. **(iii)** Investors bribe efficiently: $\Omega^{EN} = \Omega^{FB}$, income below $I - A$, not financed. **(iv)** $\Omega^{IN} = \{L \leq \rho_0\} \subset \Omega^{SB}$, financed, too much liquidation. **(v)** Small $r_b > 0$ lets the entrepreneur "buy" efficient continuation for L just above ρ_0 while investors still break even. Example ($L \sim U[0, 4]$, $U_b^0 = 1$, $\rho_1 = 4$, $\rho_0 = 2.4$, $I - A = 2.7$): cutoffs FB 3, SB 2.8 ($\mu = 1.5$), IN 2.4, IN with $r_b = 0.05$: 2.45; NPV 0.125, 0.12, 0.08, 0.0872; EN income $2.525 < 2.7$.